Zhenran Wang



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Education

Beihang University, *Ph.D. in Artificial Intelligence* Sep 2027 – Sep 2031

- **Advisor:** Prof. Feiran Huang **Research:** Artificial Intelligence, Multi-Agent Systems, Embodied Intelligence

Hefei University of Technology, *M.S. in Mathematics* Aug 2022 – Jun 2025

- **GPA:** 3.8 / 4.0 **Advisor:** Prof. Xiaofeng Chen
- **Research:** combinatorial optimization, graph theory, enumeration **Output:** two math papers — one accepted by **top-tier journal** TAMS, one under review at E-JC

Anhui University of Finance & Economics, *B.S. in Mathematics* Sep 2018 – Jul 2022

- **Core courses:** Mathematical Analysis, Advanced Algebra, Probability & Statistics, Ordinary Differential Equations, Real Analysis, Python Programming, Data Structures & Algorithms

Selected Publications (all as first or co-first author)

Embodied-SocialSim: A Device-Ready Agent Framework for Embodied Robots

Z. Wang†*, Z. Bian†, J. Li, Z. Qi* [**In Preparation**]

Builds a device-ready embodied-agent framework on Agnes Harness, connecting task state, tool approvals, execution traces, recovery, device state, sensor observations, and robot actions; integrates visual perception (e.g., YOLO), state estimation, planning, and execution, then uses a configurable simulated society to study embodied decision-making, cooperation, conflict, trust, and norm following

Visual-RSI: Can Agents Turn Visual Evidence into Better Training Data?

Z. Wang†, J. Li†, Z. Bian, F. Huang*, Z. Qi [**CVPR** In Preparation]

Designs a visual-error-driven data research framework across charts, geometry, flow diagrams, and web interfaces; plans a 7-model by 4-harness study with four diagnostic-feedback conditions. Local chart-construction checks are implemented; model gains remain unvalidated

SocietyBench: Forecasting Counterfactual Social-World Evolution

Z. Wang†*, Z. Bian†, J. Li, Z. Qi* [**AAAI 2027** Under Review]

Proposes Retrospective Evaluation: early-stage information as input, real outcomes as ground truth, entities anonymized to rule out answer recall; dual-axis calibration + timing scoring makes social forecasting objectively measurable: co-minder.github.io/Societybench

GroundBench: Multi-Resolution Polygon Grounding Exposes the Geometry Gap in Vision–Language Models

Z. Bian†*, **Z. Wang**†*, J. Li, Z. Qi* [**ICLR** Under Review]

Splits grounding into target selection and contour tracing: models pick the right object yet fail to trace its boundary; accuracy is non-monotonic in vertex budget — flat to $N{=}32$, dropping at $N{=}64$: co-minder.github.io/Groundbench

WorldCup: Prospective, Leakage-Free Evaluation of Frontier LLMs on a Live Tournament

Z. Wang†*, Z. Bian†, J. Li, Z. Qi* [**arXiv**]

Six frontier models answered before every match across 39 days — 104 matches × 7 markets, 4,494 frozen scored predictions; answers did not yet exist when asked, so leakage-freeness holds by construction, not post-hoc filtering: co-minder.github.io/worldcup2026

Hurwitz-CC: Hurwitz Numbers with Completed Cycles and Gromov–Witten Theory Relative to at Most Three Points

X. Chen, **Z. Wang** [**TAMS** Accepted]

Explicit formulas, generating functions, and structural properties for $(1, m)$-part triple Hurwitz numbers with arbitrary completed cycles: arXiv:2311.06316

Hurwitz-λ_g: Explicit Formulas and λ_g Conjecture for Double Hurwitz Numbers with Completed r-cycles

X. Chen, **Z. Wang** [**E-JC** Under Review]

First explicit formulas for one-part double Hurwitz numbers with completed r-cycles; proves a λ_g-conjecture analogue and derives the string and dilaton equations

Note: † = co-first author, ∗ = corresponding author. The mathematics papers (**TAMS**, **E-JC**) list authors alphabetically with equal contribution, per the convention in mathematics. **TAMS** = Trans. Amer. Math. Soc. (top-tier), **E-JC** = Electron. J. Combin.

Projects

RSI-Math — Failure Diagnosis and Recursive Self-Improvement for Mathematical Reasoning Models Aug 2026 – Present

- **Tech stack**: Python / PyTorch / SymPy / LLM API / Researcher Agent
- Starts from verifiable failures in algebra, geometry, combinatorics, and proof tasks; decomposes fine-grained capability gaps and generates targeted problems, solutions, counterexamples, and programmatic checks in a diagnose–intervene–retrain loop
- Starts each candidate from a common initialization M_0 and preserves generation, verification, filtering, training, and evaluation traces; uses hold-out sets, verifiers, and best-of-k controls to separate transfer from sampling or selection effects
- Tests whether repairs transfer to unseen problem types, difficulty levels, and cross-task combinations, while tracking data volume, training rounds, and verification cost for stable and reproducible recursive improvement

Embodied-SocialSim — Device-Ready Agent Framework and Social Simulation for Embodied Robots Aug 2026 – Present

- **Tech stack**: Python / Agnes Harness / ROS2 / YOLO / MHS-MOS / LLM Agent / MiroFish
- Builds a device-ready embodied-agent framework on Agnes Harness, connecting task state, tool approvals, execution traces, recovery, device state, sensor observations, and robot actions through MHS-MOS-style interfaces
- Integrates replaceable modules for visual perception (e.g., YOLO), state estimation, planning, and execution while retaining ROS2-side device interfaces; real-time motion control, interlocks, and emergency stops remain with device controllers
- Instantiates configurable robots, roles, relationships, and events as a simulated society for studying embodied perception, decision-making, cooperation, conflict, trust, and norm following—not as a project primarily about social-event forecasting

EventRadar — Multi-Platform Social-Event Collection & Processing System Jan 2026 – Jul 2026

- **Tech stack**: Python / asyncio / aiohttp / Playwright / FastAPI / SQLite / LLM API / Claude Code Skills
- End-to-end social-event intelligence: from a one-sentence topic, autonomously collects multi-platform data, runs LLM semantic processing and cross-source fusion, and outputs structured event timelines — no human in the loop
- Concurrent asyncio + aiohttp architecture covering 7 social platforms + web news, with Playwright for dynamic pages, signing service, cookie pools, proxy rotation, resumable crawls; an LLM loop (keyword generation → trial crawl → validity review) converges a one-sentence topic into $\leq$ 15 keywords with zero manual screening
- FastAPI processing service with a six-stage LLM pipeline (summarize → clean → semantic dedup → new-development extraction → compress → elaborate); news and social tracks run in parallel, then merge across sources with provenance labels (news / social / cross-verified), balancing objectivity and sentiment
- Split into **20+ Claude Code Skills** micro-workflows, intermediates persisted to SQLite for checkpoint recovery and inspection; distills **100K+ posts and 520 news articles** into ∼**30** timeline entries in one ∼40-minute run

SocietyBench — LLM Social-Event Forecasting Benchmark co-minder.github.io/Societybench Feb 2026 – Sep 2026

- **Tech stack**: Python / asyncio / FastAPI / LLM API / Claude Code Skills
- Benchmarks LLMs' social-event forecasting — not what a model knows, but whether it can, like a social researcher, anticipate an event's trajectory and public reaction from limited early information
- Retrospective Evaluation: early information of past events as input, real outcomes as ground truth, entity anonymization against "cheating"; a **dual-axis benchmark** — **probability calibration** (weighted MAE) and **temporal accuracy** (timing MAE) — each on an independent 100-point scale, plus time-decay weights and temperature control for reproducibility
- Two asyncio-parallel evaluation tracks behind a FastAPI service; ground truth auto-generated by EventRadar, so the full pipeline (anonymization → questions → parallel evaluation → report card) needs no human annotation
- Report card in ∼**1 hour**; evaluated 6 frontier models (**GPT-5.5, Gemini-3.5-Flash, Claude-Opus-4.8, DeepSeek-V4-Pro, Kimi-K2.5, Doubao-Seed-2.0-Pro**) on 125 prediction points, with a maximum single-axis gap of **21.4 points** — informing base-model selection for the in-house forecasting engine MiroFish

GroundBench — Polygon Grounding Benchmark for VLMs co-minder.github.io/Groundbench Apr 2026 – Aug 2026

- **Tech stack**: Python / PyTorch / Shapely / VLM API / RefCOCO / Claude Code Skills
- With RefCOCO-style box grounding **nearly saturated** on frontier VLMs, upgrades 1,500 image–text referring pairs from boxes to exact-N polygons under five vertex budgets, auditing region IoU and valid completion on a fixed denominator
- **Restores discrimination** where box metrics saturate: the best configuration reaches **88.2** box IoU but only **57.7** for direct polygons, non-monotonic in vertex count and collapsing at the densest budget — an "output geometry gap" in localization, boundary construction, serialization, and topology
- Evaluated 4 frontier VLMs (Doubao, Kimi, Qwen, Gemini) plus thinking/template ablations with robustness checks; vertex budgets, task forms, and models are all extensible for future geometric-grounding evaluation

LLM · WorldCup — Live LLM Forecasting on the 2026 World Cup Apr 2026 – Aug 2026

- **Tech stack**: Python / FastAPI / LLM API / GitHub Pages / Claude Code Skills
- Prospective LLM forecasting arena: 6 flagship models predict the 2026 World Cup from identical pre-match information, are scored on real results, and are ranked live at co-minder.github.io/worldcup2026
- Three task families (7 per-match markets / 12 group winners / tournament outlook), scored per market on real results with a rolling leaderboard; strict temporal isolation — models see only pre-kickoff snapshots — prevents result leakage
- Unified the 6 vendors' heterogeneous web-search interfaces and enforced one structured-prediction format across models; hard code-level gates ensure complete collection and leakage-freeness; fully scripted in Python, recomputable across sessions
- Structured profiles for all 48 teams — squads, market values, coaches, weather, venues — cross-checked across sources and updated through the tournament; three-tier raw/processed/reference storage; the frontend re-scores each round, published bilingually

Work Experience

JAC Motors, *Data Development Engineer* Jul 2025 – Apr 2026

- **Intelligent ticketing & complaint analysis**: an LLM-API complaint-handling agent for intent recognition, text classification, information extraction, and smart dispatch; FastAPI backend with Redis-cached dialogue state for multi-turn context — replaced manual triage, ∼**60%** efficiency

gain, **91%** dispatch accuracy
- Multi-source ETL pipelines (Hive + Doris) processing **tens of thousands** of records daily, end-to-end latency within 40 minutes; MySQL + Redis sustain high-concurrency reads/writes for tickets and complaints
- Agent quality-evaluation workflow with scripted test sets and metrics; prompt-strategy iteration and few-shot optimization raised classification accuracy from **76%** to **93%** and recall from **68%** to **89%**

BYD Co., Ltd., *Data & Algorithm Intern* May 2023 – Sep 2023

- **Autonomous driving — vision-only BEV 3D detection (data)**: multi-camera timestamp sync and frame alignment on an in-house annotation platform, filtering occlusion/truncation, annotation drift, and temporal anomalies; OpenCV 3D→2D projection visualization and Open3D BEV-space box validation — ∼**80K frames** quality-checked
- Multi-level BEV data augmentation (GT-paste, BEVMix, GridMask, CutOut, random flip, BEV-space random scaling); iterated with the algorithm team to gain ∼**5.2 mAP** and ∼**4.1 NDS** on the nuScenes validation set
- Automated pipeline from raw proprietary protobuf logs to nuScenes-format training data — parsing, cleaning, calibration checks, ego-pose alignment, format conversion — ∼**3**× faster data processing

Honors & Awards

Graduate Academic Scholarship, Second Class 2023
National College Mathematics Competition (CMC), Provincial Second Prize 2020
Academic Scholarship, Second Class 2019
National High School Mathematics League, Provincial Second Prize × 2 2016, 2017

Skills

- **Programming:** proficient in Python, solid in data structures & algorithms; basic C++ and MATLAB
- **AI & LLMs:** LLM-agent development; prompt engineering, multi-model API orchestration, LLM-as-a-judge evaluation; PyTorch
- **Data & engineering:** FastAPI; MySQL, Redis, SQLite; Python async programming; multi-platform data collection (Playwright, proxy rotation, account pools); message queues; modular automated pipelines
- **Engineering practice:** Git workflow and coding conventions; unit testing, logging & monitoring; Linux
- **Tools & language:** VS Code, Cursor, Claude Code; English: CET-6, strong academic reading & writing